Fiscal Year:	FY 2008	**Task Last Updated:**	FY 09/24/2009
PI Name:	Megahed, Abe B.S.		
Project Title:	Just-in-Time Simulation Platform		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	INDUSTRY	**Phone:**	(608) 827-5555
Organization Name:	Planet LLC/Orbital Technologies Corporation		
PI Address 1:	1212 Fourier Drive		
PI Address 2:			
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53717-1961	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	07/15/2008	**End Date:**	07/14/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Woolford, Barbara	**Contact Phone:**	218-483-3701
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ07JB13C		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Identification and Significance of the Innovation: Long duration spaceflight and exploration missions will require complex operations and demanding tasks. A better system for creation of intelligent simulations will be needed to prepare future crews for the rigors ahead. The software system proposed will create a method for authoring intelligent 3D simulations for training and operations support without the need for high-level programming. An authoring tool of this type will allow NASA subject matter experts to capture their knowledge and build a simulation from it, or allow developers to create scenario-based simulations for troubleshooting and repair. This tool will also be more cost effective than hardware. A specific and immediate use will be for EVA training, particularly when a EVA or IVA task needs to designed during a mission and the simulation delivered just prior to execution. • Objectives and Deliverables: Fully functional NASA Astronaut training system. After a successful tool test is complete, training materials and user-guide documentation will be completed. This will allow PLANET LLC to train a		

	number of NASA personnel on the system, and prepare the software for commercial release.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Hypercosm is currently being used by USA in its training platform known as Questus, integrated applications and tools specifically for space operations.
Task Progress:	New project for FY2008. Reporting not required for this SBIR Phase 2 project. [Ed. note: added to Task Book in September 2009 when received information on this task]
Bibliography Type:	Description: (Last Updated:)